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Paul Haskins

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Section 0.0

Tutorial 4

To understand the chirality we need to see how the state transforms under the chiral decomposition.

Definition 0.0.1: Helicity :

The projection of the spin vector onto the momentum vector. Not Lorentze invariant. Chirality is

In the massless limit the helicity is a good qm number to specify a particle. We can't flip it due to the fact that we can change the frame.

Opposite helicities $\iff$ antiparticles.

Careful when considering if we are in the massless limit.

Neutrinos are left-handed and anti-Neutrinos are right handed.

Example 0.0.1 (3.2 Peskin)

Derive the Gordon Identity

$$\bar{u}(p') \gamma^\mu u(p) = \bar{u}(p') \left[\frac{p'^\mu + p^\mu}{2m} + \frac{i\sigma^{\mu\nu} q_\nu}{2m} \right] u(p) \quad (1)$$

Use the relations

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} \mathbf{1}, \quad [\gamma^\mu, \gamma^\nu] \frac{i}{2} = \sigma^{\mu\nu}, \quad q = p' - p.$$

$$\begin{aligned} &= \frac{\bar{u}(p')}{2m} (p'^\mu + p^\mu + i\sigma^{\mu\nu} (p'_\nu - p_\nu)) u(p) \\ &= \frac{\bar{u}(p')}{2m} \left(g^{\mu\nu} p'_\mu + g^{\mu\nu} p_\mu - \frac{i}{2} [\gamma^\mu, \gamma^\nu] (p'_\nu - p_\nu) \right) u(p) \\ &= \frac{\bar{u}(p')}{2m} \left(\frac{1}{2} \{\gamma^\mu, \gamma^\nu\} (p'_\mu + p_\mu) - \frac{i}{2} [\gamma^\mu, \gamma^\nu] (p'_\nu - p_\nu) \right) u(p) \end{aligned} \quad \text{dotted}$$

(b)

$$K_0 U_{L_0} = 0.$$